Points 14/16 (87.50%)

Ex 1.

Let $K/\mathbb{Q}$ be a field extension with char 0.

Let $\alpha \in K$ with $K = F(\alpha)$ and let $A = \min_{K/F}(\alpha)$.

Let L/F be a field ext. s.t.

$$A = (T-\alpha_1)\cdot\ldots\cdot(T-\alpha_d) \text{ with } \alpha_1, \ldots, \alpha_d \in L.$$

t.s. $\Delta_{K/F}(1, \alpha, \ldots, \alpha^{d-1}) = \pm \prod_{i=1}^{d} A'(\alpha_i)$.

proof By Lemma 1.33 (i)

$$\Delta_{K/F}(1, \ldots, \alpha^{d-1}) = \det\left(\begin{pmatrix} \sigma_1(1) & \ldots & \sigma_1(\alpha^{d-1}) \\ \vdots & & \\ \sigma_d(1) & \ldots & \sigma_d(\alpha^{d-1}) \end{pmatrix}^2\right)$$

where $\sigma_i : K \to L$, pw distinct K-Hom.

From algebra 2 $\sigma_i(\alpha)$ root of A. and $\alpha_i(\alpha) \neq \alpha_j(\alpha)$ if $i \neq j$

Thus we may assume $\alpha_i(\alpha) = \alpha_i$ $\forall i$ (up to reordering)

Then.

$$A'(T) \overset{\checkmark}{=} \left(\prod_{i=1}^{d}(T-\alpha_i)\right)' \overset{\checkmark}{=} \sum_{i=1}^{d} \prod_{\substack{j=1 \\ j\neq i}}^{d} (T-\alpha_j)$$

$$A'(\alpha_k) \overset{\checkmark}{=} \sum_{i=1}^{d} \prod_{j\neq i} (\alpha_k - \alpha_j) \overset{\checkmark}{=} \prod_{j\neq k} (\alpha_k - \alpha_j)$$

$$\prod_{i=1}^{d} A'(\alpha_i) \overset{\checkmark}{=} \prod_{i=1}^{d} \prod_{j\neq i} (\alpha_i - \alpha_j) \overset{\text{tk}}{=} \prod_{\substack{i,j=1 \\ i\neq j}}^{d} (\alpha_i - \alpha_j)$$

On the other hand

$$\Delta_{K/F}(1, \alpha, \ldots, \alpha^{d-1}) = \det(\text{Vandermonde-matrix as in lecture})$$

$$\overset{\text{since}}{\checkmark} = \left(\prod_{1\leq i<j\leq d} (\alpha_j - \alpha_i)\right)^2$$

$$= \pm \prod_{\substack{i,j=1 \\ i\neq j}}^{d} (\alpha_j - \alpha_i) \overset{\checkmark}{=} \pm \prod_{i=1}^{d} A'(\alpha_i).$$

some canceled out

why?

3.5

Ex 2.

Let $K/\mathbb{Q}$ be a galois extension and $p \in \mathbb{Z}$ prime number.
We denote $P_1^{e(P_1)} \cdot \ldots \cdot P_g^{e(P_g)}$ as the prime ideal decomposition of $p\mathbb{Z}_K$, where $P_1, \ldots, P_g$ are pairwise distinct.

(i) t.s. $e(P_1) = \ldots = e(P_g)$

proof. We define $P = \{P \subseteq \mathbb{Z}_K : P \text{ an prime ideal with } P \mid p\mathbb{Z}_K\}$.
Then by Ex 158 $\mathrm{Gal}(K/\mathbb{Q}) \curvearrowright P$.
Let $i, j \in \{1, \ldots, g\}$ be distinct. By the hint from class $\mathrm{Gal}(K/\mathbb{Q})$ acts transitive, so there exists a $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$ s.t. $\sigma(P_i) = P_j$

Then it holds that

$$\prod_{k=1}^{g} P_k^{e(P_k)} = p\mathbb{Z}_K = \sigma(p\mathbb{Z}_K) = \sigma\left(\prod_{k=1}^{g} P_k^{e(P_k)}\right) = \prod_{k=1}^{g} \sigma(P_k)^{e(P_k)} \quad \text{since } \sigma \text{ an } \mathbb{Q}\text{-Automorphism}$$

hence, by choice of σ we get

$$\prod_{\substack{k=1 \\ k \neq i,j}}^{g} P_k^{e(P_k)} \cdot P_i^{e(P_i)} \cdot P_j^{e(P_j)} = \prod_{\substack{k=1 \\ k \neq i,j}}^{g} \sigma(P_k)^{e(P_k)} \cdot \underbrace{\sigma(P_i)^{e(P_i)}}_{= P_j^{e(P_i)}} \cdot \sigma(P_j)^{e(P_j)} = P_i^{e(P_j)}$$

By the uniqueness of the prime ideal decomposition (up to ordering) we get that $\prod_{\substack{k=1 \\ k \neq i,j}}^{g} P_k^{e(P_k)} = \prod_{\substack{k=1 \\ k \neq i,j}}^{g} \sigma(P_k)^{e(P_k)}$

Thus $P_i^{e(P_i)} = \sigma\left(P_j^{e(P_j)}\right) = P_i^{e(P_j)}$ and $P_j^{e(P_j)} = \sigma\left(P_i^{e(P_i)}\right) = P_i^{e(P_i)}$

and therefore $e(P_i) > \frac{1}{2} e(P_j)$. Since $i \neq j \in \{1, \ldots, g\}$ were arbitrary we can conclude. ∎ (1.5)

(ii) t.s. $f(P_1) = \ldots = f(P_g)$

proof. By definition $N(P_k) = p^{f(P_k)}$ $\forall k \in \{1, \ldots, g\}$.

Then

$$p^{[K:\mathbb{Q}]} \overset{\checkmark}{=} N(p\mathbb{Z}_K) \overset{\checkmark}{=} N\Big(\prod_{k=1}^{g} P_i^{\overbrace{e(P_k)}^{= e(P_i) \text{ by (i)}}}\Big) \overset{\checkmark}{=} \prod_{k=1}^{g} N(P_k)^{e(P_k)} \overset{\checkmark}{=} \prod_{k=1}^{g} p^{f(P_k)e(P_k)}$$

Thus $p^{\frac{[K:\mathbb{Q}]}{e(P_i)}} = \prod_{i=1}^{g} p^{f(P_i)}$ ok

Let $i, j \in \{1, \ldots, g\}$ be distinct. Then

$$p^{f(P_i)} \cdot \prod_{\substack{k=1 \\ k \neq i}}^{g} p^{f(P_k)} \overset{\text{ok}}{=} p^{\frac{[K:\mathbb{Q}]}{e(P_i)}} \overset{\text{ok}}{=} \left(\prod_{\substack{k=1 \\ k \neq j}}^{g} p^{f(\sigma(P_k))}\right) \cdot p^{f(P_j)}$$

why? thus $\prod_{\substack{k=1 \\ k \neq i}}^{g} p^{f(P_k)} = \prod_{\substack{k=1 \\ k \neq j}}^{g} p^{f(\sigma(P_k))}$

By the same argument it follows that $p^{f(P_i)} = p^{f(P_j)}$ thus $f(P_i) = f(P_j)$
and we can conclude thus $f(P_1) = \ldots = f(P_g)$ ∎ (1.5)

Ex 2.

(ii)

t.s. $\forall i \in \{1, \dots, g\}: e(P_i) \cdot f(P_i) g = [K : \mathbb{Q}]$.

proof. Let $i \in \{1, \dots, g\}$. By Lemma 3.7 it holds that

$$\sum_{k=1}^{g} e(P_k) f(P_k) \overset{\checkmark}{=} [K : \mathbb{Q}]$$

$e(P_i) f(P_i)$ by (i), (ii)

$$\overset{\checkmark}{=} e(P_i) f(P_i) \cdot g$$

Thus $e(P_i) \cdot f(P_i) g = [K : \mathbb{Q}]$. ∎ (1)

Exercise 3

Let $p \in \mathbb{Z}$ prime number, $K = \mathbb{Q}(\zeta)$, $\zeta = \exp\left(\frac{2\pi i}{p}\right)$.

Let $\ell \in \mathbb{Z}$ be a prime number, s.t. $\ell \neq p$.

(v) t.s. $\ell \mathbb{Z}_K = P_1 \cdot \ldots \cdot P_g$, where $P_1, \ldots, P_g \subseteq \mathbb{Z}_K$ pairwise distinct prime ideal with $f(P_i) = f$, $e(P_i) = 1$, $g = \frac{p-1}{f}$

proof. Since $\mathbb{Z}_K$ is dedekind

$$\ell \mathbb{Z}_K = P_1^{e(P_1)} \cdot \ldots \cdot P_g^{e(P_g)}, \text{ where } P_i \subseteq \mathbb{Z}_K \text{ prime ideals.}$$

Thm by ex 02, $e(P_1) = \ldots = e(P_g)$, $f(P_1) = \ldots = f(P_g) \overset{\text{sheet 08}}{=} f$ ✓

and

$$[K:\mathbb{Q}] \overset{\text{ex 02}}{=} \sum_{1} e(P_i) f(P_i) = e(P_i) f g \quad \forall i$$

Since $\min(\zeta) = 1 + \ldots + X^{p-1} \Rightarrow [K:\mathbb{Q}] = p-1$.

Since $\ell \neq p$, Corollary 3.27 implies that $e(P_i) = 1$

So we can conclude

$$p-1 = [K:\mathbb{Q}] = fg \Rightarrow g = \frac{[K:\mathbb{Q}]}{f} = \frac{p-1}{f}. \quad \square$$

Ex 4.

Let $p \in \mathbb{Z}$ be a positive prime number and Φ_p the p-th cyclotomic polynomial.

i) Let $a \in \mathbb{Z}$ and $\ell \in \mathbb{Z}$ prime s.t. $\ell \mid \Phi_p(a)$ and $\ell \neq p$.

t.s. $\ell \equiv 1 \pmod p$

proof From alg 2 we know $\Phi_p = 1 + \dots + X^{p-1} \in \mathbb{Z}[X]$ thus $\Phi_p(X-1) = X^p - 1$. (is this a product?)

Thus $\ell \mid \Phi_p(a)$ implies that $\ell \mid a^p - 1$ and therefore $a^p - 1 \in (\ell)$.

In particular, it holds that $a \neq 1$. otherwise $\ell \mid 1 + \dots + 1^{p-1} = p$ which is absurd since p, ℓ distinct prime. Then

$$0 = \pi_\ell(a^p - 1) = \pi_\ell^p(a) - \underbrace{\pi_\ell(1)}_{=1 \text{ in } \mathbb{Z}/\ell\mathbb{Z}}$$

where $\pi_\ell : \mathbb{Z} \to \mathbb{Z}/\ell\mathbb{Z}$ canon. proj.

Then it holds that $\langle \pi(a) \rangle \leq \left(\mathbb{Z}/\ell\mathbb{Z}\right)^*$, where $\langle \pi_\ell(a) \rangle = \{\pi_\ell^m(a) \mid m \in \mathbb{N}\}$, since $\pi^p(a) = 1$.

claim There exists no $n \in \mathbb{N}$ s.t. $\pi_\ell^n(a) = 1$ in $\mathbb{Z}/\ell\mathbb{Z}$. → written in this way the claim is wrong

By claim we get $p = \#\langle \pi(a) \rangle$ and Lagrange provides us that

$$p \mid \#\left(\mathbb{Z}/\ell\mathbb{Z}\right)^* = \ell - 1 \text{ thus } (-1) \in (p) \Rightarrow \ell \equiv 1 \pmod p.$$

it is left to prove claim.

Assuming there exists $n \in \mathbb{N}$ s.t. $\pi_\ell^n(a) = 1$, there must be a minimal $\tilde{n} \in \mathbb{N}$ with $\pi_\ell^{\tilde{n}}(a) = 1$ s.t. $\tilde{n} \mid p$. But since p prime we get $\tilde{n} = 1$ but then $\pi_\ell(a) = 1$ in $\mathbb{Z}/\ell\mathbb{Z}$ thus $a = 1$ which is absurd. ☐ 1.75

ii) Let $\ell_1, \dots, \ell_n$ prime numbers with $n \geq 0$.

t.s. $\exists\, \ell \in \mathbb{Z}$ prime with $\ell \notin \{\ell_1, \dots, \ell_n\}$ and $\ell \equiv 1 \pmod p$.

proof We fix $a = \ell_1 \cdot \dots \cdot \ell_n \in \mathbb{Z}$. ok

Then $\forall i \in \{1, \dots, n\} : \ell_i \nmid \Phi_p(a) = 1 + \sum_{k=1}^{p-1} a^k \quad (*)$

(otherwise $\ell_i \mid \Phi_p(a)$, $\ell_i \mid \sum_{k=1}^{p-1} a^k \Rightarrow 1 = \Phi_p(a) - \sum_{k=1}^{p-1} a^k \in (\ell_i)$ but ℓ_i prime) ✓

But since $\mathbb{Z}$ factorial and $\Phi_p(a)$ positive ($\ell_i > 1$), there exists a $\ell \in \mathbb{Z}$ prime s.t. $\ell \mid \Phi_p(a)$. ✓

Then we claim that $\ell \neq p$. Indeed, if $\ell = p$,

it follows that $p \mid \Phi_p(a) \cdot (a-1) = a^p - 1$. But then $a^p \equiv 1 \bmod (p)$

and $p \nmid a$ (otherwise $p \mid \ell_i$ but $\ell_i \nmid \Phi_p(a)$ ↯) ↯

so a corollary of Fermat's little theorem implies that $a^{p-1} \equiv 1 \pmod p$

Together with $a^p \equiv 1 \bmod (p)$ this gives us $a = \ell_1 \cdot \dots \cdot \ell_n \equiv 1 \pmod p$ why? ??

thus $a \in (p)$. But this contradicts the factoriality of $\mathbb{Z}$. ???

Therefore $\ell \neq p$. So ℓ fulfills the assumption of (i) and we get that

$$\ell \equiv 1 \pmod p$$

But since $\ell \mid \Phi_p(a)$ it holds that $\ell \notin \{\ell_1, \dots, \ell_n\}$ and we can conclude.

☐

What's the difference between $(\bmod p)$ and $\bmod (p)$?

The argument is perfect. The part where you check $\ell \neq p$ doesn't convince me.

1.75

Ex 4.

(iii)

t.s. $L=\{\ell \in \mathbb{Z} \mid \ell \text{ prime number s.t } \ell \equiv 1 \pmod p\}$ is infinite.

proof. We assume that L is finite. So we can write

$L = \{\ell_1, \dots, \ell_n\}$, for $\ell_i \in \mathbb{Z}$ prime number with $\ell_i \equiv 1 \pmod p$ for every $i \in \{1, \dots, n\}$. and for a $n > 0$.

But by (ii) we can find a prime number $\ell \in \mathbb{Z}$ s.t. $\ell \notin L$ and $\ell \equiv 1 \pmod p$ which is by assumption a contradiction and we can conclude.

□

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